

HUW CHESTON

Website: <https://huwcheston.github.io/>

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EXPERIENCE

Data Scientist

July 2025 –

Colossal Sound (Part-Time)

Remote

- Conducted machine learning research and development for a fast-growing UK startup operating at the intersection of **music recommendation and hip-hop**.

Research Assistant, Multimodal AI

May 2025 –

Centre for Digital Music, Queen Mary University of London (Part-Time)

London, UK

- Conducted research into multimodal machine perception **in collaboration with Meta** (Project Aria).
- Developed machine-learning models that combine vision and audio to produce representations of real-world scenes.

Research Scientist Intern, Audio Intelligence

June – August 2024

Spotify

London, UK

- Developed end-to-end deep learning model for identifying copyrighted samples in catalogue audio recordings, **driving downstream product applications** in music content categorisation & plagiarism detection
- Results **outperformed competing system (“Shazam”) by 9x**, improved upon internal model by **13%**
- Managed pipelines for generating artificial training data at **petabyte-scale** using Google Cloud & Apache Beam
- Deployed training runs on distributed GPU clusters using Ray & Kubernetes
- Presented results** to senior company stakeholders and in a scientific paper [accessible at [arXiv:2502.06364](https://arxiv.org/abs/2502.06364)]

Music Data Science Lecturer & Supervisor

2021 – 2024

University of Cambridge

Cambridge, UK

- Delivered **100+** undergraduate supervisions and lectures on modelling and visualisation of audio data
- Managed** 4 undergraduate student dissertations on music-related data science and machine learning projects

Teaching Assistant

2020 – 2021

Kingswood School

Bath, UK

- Planned & taught music technology lessons, both in-classroom and virtually during COVID
- Managed recording studio and music technology suite, produced promotional audio-visual material for the school

Professional Musician

2016 – 2020

Freelance

UK

- Worked with internationally recognised acts including *Clean Bandit*, *Everything Everything*, & *Dinosaur*.
- Performed regularly around the UK & abroad, in front of crowds exceeding 30,000+ capacity.

EDUCATION

University of Cambridge

2021 – 2025

Ph.D, Music Computing

Cambridge, UK

- Advanced state-of-the-art** in music computing tasks including automatic artist identification from audio
- Published **5 first author** papers with several in major (top-20) science & computer science journals
- Fully-funded with £80k competitive research grant from Cambridge Trust
- Service:** peer review for *Transactions of the International Society of Music Information Retrieval*, *Music & Science* journals, *International Society for Music Information Retrieval* conference

University of Oxford

2016 – 2020

BA + MSt., Music

Oxford, UK

- Graduated with **highest mark in year**, 85% average
- Fully-funded masters study with £25k research grant from Linacre College, Oxford

SKILLS

Stats: multi-level modelling, NHST workflows, time-series analysis, experiment design, optimisation
Inference: simulation, AB testing, hypothesis testing, bootstrapping, causal inference, dimensionality reduction,
Machine Learning: explainability, generative modelling, artificial datasets, neural networks, big data
Languages: Python, JavaScript, HTML/CSS, R, SQL
Tools: Git, Unix, L^AT_EX, Google Cloud Platform, Apache Beam, Docker, Ray, MLFlow
Libraries: pandas, matplotlib, scipy, scikit-learn, pytorch, statsmodels, huggingface, numpy, librosa, captum, open-ai
Domain Knowledge: music categorisation & retrieval, audio signal processing, natural language processing
Audio Production: ambisonics, binaural audio, multichannel microphone array setups, digital audio workstations

OPEN SOURCE CONTRIBUTIONS

mirdata (190k+ installs) major feature development and optimisations to data architecture for a Python library that standardizes access to audio datasets commonly used in machine learning research and development
miditok (150k+ installs) bug fixes and optimisations for a music tokenization Python library used in generative AI music

SELECTED PUBLICATIONS

Journal Articles

- Cheston, H., Cross, I., & Harrison, P. M. C. (2024). Trade-offs in coordination strategies for duet jazz performances subject to network delay and jitter. *Music Perception*, 42(1), 48–72.
<https://doi.org/10.31234/osf.io/z8c7w>
- Cheston, H., Schlichting, J. L., Cross, I., & Harrison, P. M. C. (2024a). Jazz trio database: Automated annotation of jazz piano trio recordings processed using audio source separation. *Transactions of the International Society for Music Information Retrieval*, 7(1), 144–158.
<https://doi.org/10.5334/tismir.186>
- Cheston, H., Schlichting, J. L., Cross, I., & Harrison, P. M. C. (2024b). Rhythmic qualities of jazz improvisation predict performer identity and style in source-separated audio recordings. *Royal Society Open Science*, 11(11), 240920. <https://doi.org/10.1098/rsos.240920>

Preprints

- Cheston, H., Balen, J. V., & Durand, S. (2025). Automatic identification of samples in hip-hop music via multi-loss training and an artificial dataset. *arXiv*. <https://doi.org/10.48550/arXiv.2502.06364>
- Cheston, H., Bance, R., & Harrison, P. M. C. (2025). Deconstructing jazz piano style using machine learning. *arXiv*. <https://doi.org/10.48550/arXiv.2504.05009>

Conference Proceedings

- Cheston, H., Bance, R., & Harrison, P. (2024). Characterizing jazz improvisation style through explainable performer identification models. *DMRN+19: Digital Music Research Network Workshop*.
<https://www.qmul.ac.uk/dmrn/media/dmrn/DMRN-19-proceedings--January.pdf>
- Cheston, H., Cross, I., & Harrison, P. (2023). An automated pipeline for characterizing timing in jazz trios. *DMRN+18: Digital Music Research Network Workshop*.
<https://www.qmul.ac.uk/dmrn/media/dmrn/DMRN-18-Proceedings.pdf>

REFERENCES AVAILABLE ON REQUEST
